App Dev2 Project Report

Trekking Management System: TrekMate

1. Student Details

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About me :

I am a passionate and dedicated software developer with a strong interest in building innovative and user-friendly web applications.I enjoy learning new technologies, solving real-world problems, and continuously improving my technical skills.My goal is to create impactful digital solutions while growing as a full-stack developer.

2. About Me

I am currently pursuing the BS Degree in Data Science and Applications at the Indian Institute of Technology Madras (IIT Madras). I am passionate about full-stack web development and enjoy building practical, user-focused applications using modern technologies.

3. Project Details

Project Title: TrekMate – Trekking Management Application

Problem Statement:
To design and develop a web-based trekking management application that enables users to discover, book, and manage trekking expeditions while providing administrators and trek staff with an efficient platform to organize treks, manage bookings, assign staff, and monitor trekking activities.

Approach:
The application was developed using Flask as the backend framework and Vue.js for the frontend, following a modular architecture. It provides role-based access for Admin, Trek Staff, and Users, allowing administrators to manage treks and bookings, staff to handle assigned expeditions, and users to browse, book, and track their trekking journeys through an interactive and user-friendly interface.

4. Technologies and Frameworks used

Technology / Library	Purpose
Flask	Core backend framework for building REST APIs
Vue.js	Frontend framework for developing an interactive single-page application
SQLAlchemy	Object Relational Mapper (ORM) for database operations
SQLite	Lightweight relational database for storing application data
JWT (Flask-JWT-Extended)	Secure user authentication and role-based authorization
Bootstrap 5	Responsive UI design and frontend styling
Axios	Handling HTTP requests between Vue.js frontend and Flask backend
Redis	Caching frequently accessed data and supporting background task processing
Celery	Executing asynchronous and scheduled background tasks
Jinja2	Rendering the initial HTML template for the Flask application entry point
Chart.js	Displaying dashboard analytics and summary charts
HTML5, CSS3 & JavaScript (ES6)	Building responsive layouts, styling, and client-side functionality
Python	Primary programming language for backend development

5. Database Schema / ER Diagram

Tables:

User – stores user, admin, and staff account information including profile details, login credentials, contact information, role, and account status.

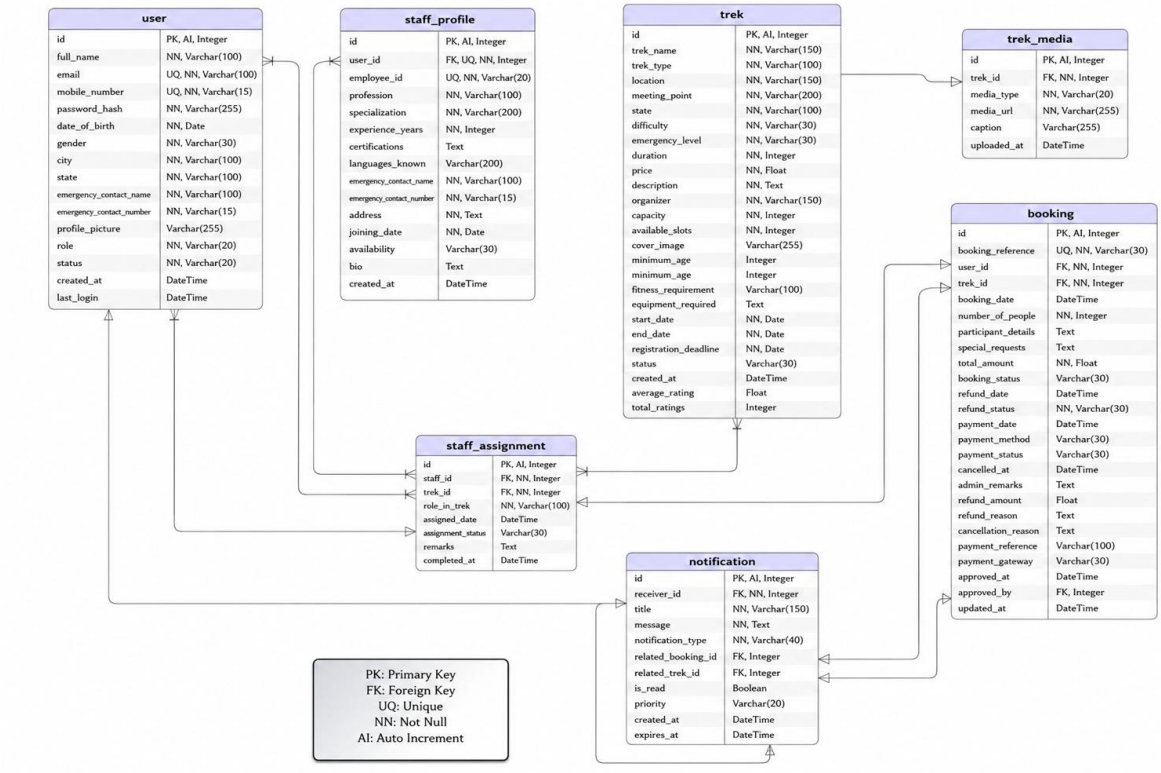
Trek – stores trekking expedition information such as trek type, location, schedule, difficulty level, pricing, capacity, and availability.

Booking – stores trek booking details including participants, payment information, booking status, cancellation, and refund records.

StaffProfile – stores additional professional information of trekking staff including specialization, experience, certifications, languages, and availability.
StaffAssignment – stores staff assignments for treks, including assigned role, assignment status, remarks, and completion details.
TrekMedia – stores trek-related images and videos along with media type, captions, and upload information.
Notification – stores system notifications sent to users regarding bookings, treks, payments, approvals, and other important updates.

Relationships:

- **One-to-Many → User → Booking**
- **One-to-One → User → StaffProfile**
- **One-to-Many → User (Staff) → StaffAssignment**
- **One-to-Many → User → Notification**
- **One-to-Many → User (Admin) → Booking (Approved By)**
- **One-to-Many → Trek → Booking**
- **One-to-Many → Trek → TrekMedia**
- **One-to-Many → Trek → StaffAssignment**
- **One-to-Many → Trek → Notification**
- **One-to-Many → Booking → Notification**



6. API Resource Endpoints

Endpoint	Method	Description
/api/register	POST	Register a new user account
/api/login	POST	Authenticate user and generate JWT token
/api/admin/dashboard	GET	Retrieve admin dashboard statistics
/api/admin/treks	GET	Fetch all trekking expeditions
/api/admin/create-trek	POST	Create a new trekking expedition
/api/admin/edit-trek/{trek_id}	PUT	Update trek information
/api/admin/delete-trek/{trek_id}	DELETE	Delete a trek from the system
/api/admin/bookings	GET	Retrieve all trek bookings
/api/admin/assign-staff	POST	Assign staff members to a trek
/api/user/dashboard	GET	Retrieve user dashboard information
/api/user/profile	GET	Fetch logged-in user's profile
/api/user/browse-treks	GET	Display all available trekking expeditions
/api/user/book-trek/{trek_id}	POST	Book a trekking expedition
/api/user/my-bookings	GET	Retrieve user's booking history
/api/staff/dashboard	GET	Retrieve staff dashboard details

Endpoint	Method	Description
/api/staff/assigned-treks	GET	Fetch all treks assigned to a staff member
/api/staff/participants/{trek_id}	GET	View participants of an assigned trek
/api/staff/complete-trek/{assignment_id}	PUT	Mark an assigned trek as completed

7. Architecture and Feature

Architecture Overview:

- app.py – Main Flask application entry point and application initialization.
- models.py – Database models implemented using SQLAlchemy ORM.
- routes.py – Flask Blueprint containing REST API endpoints for Admin, User, and Staff modules.
- celery_config.py – Configuration for Celery background task processing.
- tasks.py – Background tasks for notifications, reminders, and scheduled jobs.
- schedule.py – Periodic task scheduling using Celery Beat.
- mails.py – Email service for booking confirmations and notifications.
- templates/ – Jinja2 templates used for the Flask entry point.
- static/ – Stores uploaded trek images, profile pictures, CSS, and other static assets.
- frontend/src/ – Vue.js components, views, routing, and frontend assets.

Implemented Features:

- User registration and JWT-based authentication
- Role-based access control (Admin, Staff, and User)
- Trek creation, update, and management
- Trek browsing with search and filtering
- Online trek booking and booking management
- Staff assignment and trek completion workflow
- User, Admin, and Staff dashboards
- Profile management with image upload
- Trek image gallery and media management
- Booking approval, rejection, and cancellation
- Redis-based caching for improved performance
- Celery background tasks and scheduled email notifications
- Responsive user interface developed using Vue.js and Bootstrap 5

Additional Features:

- Trek popularity and recommendation section

- Responsive landing page with gallery and testimonials
- Secure password hashing and JWT authentication
- SQLite database with SQLAlchemy ORM
- Image upload support for treks and user profiles
- Dynamic dashboards with interactive charts using Chart.js

8. Google Drive Video Link

https://drive.google.com/file/d/1UW7ZTg-1oo8lQF14narAZ1ubsnULlB71/view?usp=sharing